Agent Task Specification

KV-cache quantization benchmark - full-precision (f16) vs quantized (q8_0, turbo4). Deterministic data, seed=42. All answers auto-computed.

Purpose

Each task is run through the agent loop (model + tools over multiple turns) for every KV-cache config, N trials each. The model must USE the tools; the harness executes them and feeds results back. Success is an exact match to the expected answer. Difficulty scales with context length and step count so that the full-precision model succeeds while the quantized model starts to fail - that gap is the result.

Tools available to the agent

- calculator(expression): evaluates arithmetic, e.g. '(68 + 83) * 2'. Returns a number.
- lookup(key): returns the value of a named fact. ONLY provided for Section C. In Section A the facts are in the prompt (the agent must recall them from context, not look them up).

Scoring

Per trial: success = the final number equals 'expected' exactly. Report per config: success rate, avg steps, tool-call validity, avg latency, and pass^k consistency across trials.

System prompt (same for every task)

```
You are a precise tool-using assistant. Use the calculator tool for all
arithmetic and do not guess. When you have the final number, reply with
just that number and nothing else.
```

Section A - Buried-fact recall (directly stresses KV cache)

The agent is given a long list of sensor readings IN THE PROMPT, told to do one calculator step (a distraction that pushes the target far back in context), then asked to report one specific reading from earlier. No lookup tool - it must recall from the KV cache. Longer list = more cache pressure. This is the sharpest test of KV-cache compression.

Task recall_40 (40 readings) Expected: 45

Data given in the prompt (remember these):

```
sensor_01: 91   sensor_02: 24   sensor_03: 13   sensor_04: 45   sensor_05: 41   sensor_06: 38
sensor_07: 27   sensor_08: 23   sensor_09: 96   sensor_10: 79   sensor_11: 21   sensor_12: 85
sensor_13: 64   sensor_14: 14   sensor_15: 13   sensor_16: 21   sensor_17: 37   sensor_18: 39
sensor_19: 74   sensor_20: 87   sensor_21: 13   sensor_22: 81   sensor_23: 35   sensor_24: 93
sensor_25: 99   sensor_26: 79   sensor_27: 63   sensor_28: 38   sensor_29: 67   sensor_30: 85
sensor_31: 45   sensor_32: 10   sensor_33: 30   sensor_34: 99   sensor_35: 64   sensor_36: 53
sensor_37: 45   sensor_38: 29   sensor_39: 37   sensor_40: 53
```

Prompt to the agent:

```
Here are 40 sensor readings (above). Using them:
1. Use the calculator to add sensor_39 and sensor_40 together.
2. Then tell me the exact value of sensor_04.
Reply with just the value of sensor_04.
```

(Distraction step sensor_39+sensor_40 = 90. Correct answer is sensor_04 = 45.)

Task recall_80 (80 readings) Expected: 87

Data given in the prompt (remember these):

```
sensor_01: 23   sensor_02: 21   sensor_03: 58   sensor_04: 22   sensor_05: 55   sensor_06: 54
sensor_07: 87   sensor_08: 43   sensor_09: 15   sensor_10: 68   sensor_11: 78   sensor_12: 25
sensor_13: 58   sensor_14: 20   sensor_15: 80   sensor_16: 47   sensor_17: 90   sensor_18: 89
sensor_19: 56   sensor_20: 83   sensor_21: 34   sensor_22: 18   sensor_23: 15   sensor_24: 94
sensor_25: 39   sensor_26: 47   sensor_27: 20   sensor_28: 39   sensor_29: 22   sensor_30: 58
sensor_31: 45   sensor_32: 68   sensor_33: 91   sensor_34: 56   sensor_35: 30   sensor_36: 57
sensor_37: 55   sensor_38: 36   sensor_39: 95   sensor_40: 44   sensor_41: 99   sensor_42: 97
sensor_43: 92   sensor_44: 19   sensor_45: 87   sensor_46: 91   sensor_47: 31   sensor_48: 78
sensor_49: 41   sensor_50: 30   sensor_51: 69   sensor_52: 58   sensor_53: 44   sensor_54: 91
sensor_55: 98   sensor_56: 81   sensor_57: 38   sensor_58: 97   sensor_59: 51   sensor_60: 17
sensor_61: 39   sensor_62: 14   sensor_63: 50   sensor_64: 61   sensor_65: 44   sensor_66: 18
sensor_67: 37   sensor_68: 82   sensor_69: 50   sensor_70: 37   sensor_71: 93   sensor_72: 73
sensor_73: 60   sensor_74: 92   sensor_75: 68   sensor_76: 28   sensor_77: 43   sensor_78: 27
sensor_79: 41   sensor_80: 81
```

Prompt to the agent:

```
Here are 80 sensor readings (above). Using them:
1. Use the calculator to add sensor_79 and sensor_80 together.
2. Then tell me the exact value of sensor_07.
Reply with just the value of sensor_07.
```

(Distraction step sensor_79+sensor_80 = 122. Correct answer is sensor_07 = 87.)

Task recall_120 (120 readings) Expected: 73

Data given in the prompt (remember these):

```
sensor_01: 78   sensor_02: 43   sensor_03: 84   sensor_04: 64   sensor_05: 84   sensor_06: 61
sensor_07: 56   sensor_08: 38   sensor_09: 27   sensor_10: 75   sensor_11: 73   sensor_12: 21
sensor_13: 16   sensor_14: 24   sensor_15: 29   sensor_16: 90   sensor_17: 30   sensor_18: 97
sensor_19: 64   sensor_20: 86   sensor_21: 18   sensor_22: 59   sensor_23: 58   sensor_24: 86
sensor_25: 69   sensor_26: 77   sensor_27: 42   sensor_28: 80   sensor_29: 11   sensor_30: 97
sensor_31: 24   sensor_32: 97   sensor_33: 78   sensor_34: 44   sensor_35: 92   sensor_36: 53
sensor_37: 24   sensor_38: 47   sensor_39: 65   sensor_40: 30   sensor_41: 68   sensor_42: 10
sensor_43: 43   sensor_44: 74   sensor_45: 32   sensor_46: 74   sensor_47: 23   sensor_48: 90
sensor_49: 48   sensor_50: 91   sensor_51: 74   sensor_52: 87   sensor_53: 35   sensor_54: 29
sensor_55: 57   sensor_56: 30   sensor_57: 79   sensor_58: 77   sensor_59: 10   sensor_60: 86
sensor_61: 51   sensor_62: 72   sensor_63: 12   sensor_64: 24   sensor_65: 56   sensor_66: 49
sensor_67: 40   sensor_68: 17   sensor_69: 40   sensor_70: 82   sensor_71: 20   sensor_72: 20
```

```
sensor_73: 72   sensor_74: 18   sensor_75: 78   sensor_76: 26   sensor_77: 26   sensor_78: 94
sensor_79: 70   sensor_80: 80   sensor_81: 31   sensor_82: 43   sensor_83: 77   sensor_84: 87
sensor_85: 64   sensor_86: 37   sensor_87: 79   sensor_88: 98   sensor_89: 35   sensor_90: 49
sensor_91: 61   sensor_92: 95   sensor_93: 93   sensor_94: 57   sensor_95: 66   sensor_96: 76
sensor_97: 67   sensor_98: 25   sensor_99: 41   sensor_100: 38   sensor_101: 18   sensor_102: 53
sensor_103: 12   sensor_104: 85   sensor_105: 80   sensor_106: 39   sensor_107: 85   sensor_108: 38
sensor_109: 10   sensor_110: 19   sensor_111: 90   sensor_112: 17   sensor_113: 39   sensor_114: 18
sensor_115: 14   sensor_116: 52   sensor_117: 19   sensor_118: 75   sensor_119: 40   sensor_120: 45
```

Prompt to the agent:

```
Here are 120 sensor readings (above). Using them:
1. Use the calculator to add sensor_119 and sensor_120 together.
2. Then tell me the exact value of sensor_11.
Reply with just the value of sensor_11.
```

(Distraction step sensor_119+sensor_120 = 85. Correct answer is sensor_11 = 73.)

Section B - Running-state tracking (precision over a long trajectory)

The agent applies a sequence of signed transactions to a running balance, one calculator step each, and reports the final total. Each step appends to the context, so the trajectory grows - quantized cache drifts and loses the exact total. Longer sequence = harder.

Task balance_8 (8 transactions) Expected: -111

Prompt to the agent:

```
You are tracking a balance. Starting balance is 0. Apply these
8 transactions in order, using the calculator for each step,
keeping the running balance:
-119, +252, +252, -107, +59, -191, -220, -37
Reply with just the final balance.
```

(Correct running total = -111.)

Task balance_12 (12 transactions) Expected: -384

Prompt to the agent:

```
You are tracking a balance. Starting balance is 0. Apply these
12 transactions in order, using the calculator for each step,
keeping the running balance:
+41, -183, +137, +107, -81, -103, -246, +48, -291, +35, +57, +95
Reply with just the final balance.
```

(Correct running total = -384.)

Task balance_16 (16 transactions) Expected: -235

Prompt to the agent:

```
You are tracking a balance. Starting balance is 0. Apply these
16 transactions in order, using the calculator for each step,
keeping the running balance:
-258, -119, -40, +204, +209, -242, -226, -89, +161, +39, +170, +35, -267, +39, +105, +44
Reply with just the final balance.
```

(Correct running total = -235.)

Task balance_20 (20 transactions) Expected: 180

Prompt to the agent:

```
You are tracking a balance. Starting balance is 0. Apply these
20 transactions in order, using the calculator for each step,
keeping the running balance:
+216, +136, +51, -299, -143, +170, +145, -77, -244, -47, +244, +47, +269, -77, -45, +199, -90, -288,
-280, +293
Reply with just the final balance.
```

(Correct running total = 180.)

Section C - Multi-hop chain (error compounding)

The agent looks up each value (lookup tool IS provided here) and performs a chain of dependent operations. One slip in any step ruins the final answer. More hops = more chances to compound an error.

Task chain3 Expected: 250

Facts (retrievable via the lookup tool):

```
base: 240   factor: 5   offset: 200   divisor: 4
```

Prompt to the agent:

```
Compute (base * factor - offset) / divisor. Look up each value with the lookup tool as you
need it and use the calculator for each operation.
Reply with just the final number.
```

(Correct answer = 250.)

Task chain4 Expected: 396

Facts (retrievable via the lookup tool):

```
p: 18   q: 7   r: 6   s: 3
```

Prompt to the agent:

```
Compute (p * q + r) * s. Look up each value with the lookup tool as you
need it and use the calculator for each operation.
Reply with just the final number.
```

(Correct answer = 396.)

Task chain5 Expected: 3345

Facts (retrievable via the lookup tool):

```
a: 100   b: 4   c: 9   d: 2   e: 15
```

Prompt to the agent:

```
Compute ((a / b) * c - d) * e. Look up each value with the lookup tool as you
need it and use the calculator for each operation.
Reply with just the final number.
```

(Correct answer = 3345.)

Summary of all tasks

```
SECTION A - recall
recall_40            40 readings              = 45
recall_80            80 readings              = 87
recall_120           120 readings             = 73
SECTION B - balance
balance_8            8 txns                   = -111
balance_12           12 txns                  = -384
balance_16           16 txns                  = -235
balance_20           20 txns                  = 180
SECTION C - multi-hop
chain3               (base * factor - offset) / divisor = 250
chain4               (p * q + r) * s          = 396
chain5               ((a / b) * c - d) * e    = 3345
```

Total: 3 recall + 4 state + 3 multi-hop = 10 tasks. Run each at 5 trials per KV config.